

Small swaps force a lower bound

Name: _____

Date: _____

New clue type: an entry clue fixes the height **in one cell**. There are no outside visibility clues. Rows and columns still contain 1–4 exactly once.

1. A switch that preserves every rule.

Copy this Latin square. In the upper-left 2-by-2 block, exchange the 1s and 2s. Why is the new square still legal?

1	2	3	4
2	1	4	3
3	4	1	2
4	3	2	1

2. Four independent hiding places.

Each bold 2-by-2 block has a similar switch. Suppose you reveal at most three entry clues from the original square. Show that a second completion always remains.

Why must one whole switch escape the clues?

3. Is touching every block sufficient?

Reveal just cells (row 1, column 1), (1,3), (3,1), (3,3). They touch all four bold blocks. Can you still make another square?

An alternative completion or a convincing reason:

Go further: A *Latin trade* changes some entries and preserves every row and column. Explain why a uniquely determining entry set must touch every possible trade, not merely these four.

Build first. Explain by pointing, talking, or drawing.